



Technical Note

Data integrity resolving gas-solid interfacial heat transfer coefficient in porous media under low Reynolds number

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ABSTRACT

Inverse computation of gas-solid interfacial heat transfer coefficient (h_{sg}) in porous media has remained a challenging issue. This work proposes a method using data integrity of both solid (T_s) and gas (T_g) temperatures in porous media to resolve this issue. The data integrity is accomplished by simultaneous measurement of T_s and T_g developed in our previous work. h_{sg} is approximately solved by the discretized gas energy conservation equation in a central difference scheme implemented with data integrity of T_s and T_g . This solution takes advantages in reducing both computation cost and numerical uncertainty since only one unknown parameter is directly calculated from one equation. An empirical correlation of $Nu = 0.001(Re^{1.66}Pr^{1/3})$ ($R^2 = 0.93$) is established for low- Re gas flow in porous media ($5 \text{ mm} \leq d_p \leq 15 \text{ mm}$ and $13 \leq Re \leq 133$) based on in total 54 studied cases considering four variables of Darcy flux, particle size, porous medium, and heating temperature. The proposed method is validated by the consistent Nu correlations in the literature. This work gives a new perspective of data integrity to resolve the challenging inverse models for heat transfer in porous media.

1. Introduction

Gas-solid heat transfer in porous media under low Reynolds (Re) number ($\leq O(10^2)$) conditions is an important transport phenomenon across energy, environmental and chemical engineering applications [1–3]. Gas flow in porous media under low- Re conditions exhibits a viscosity-dominated regime, in which the flow boundary layer could be significant thickened [4]. In this regime, the convective heat transfer reduces due to small flow velocities, while the conductive heat transfer becomes relatively more significant, particularly in the gas phase, which dominates the overall thermal resistance. The low thermal conductivity of gas phase leads to slower thermal response compared to the solid phase, resulting in temperature gradients at the gas-solid interface. This phenomenon renders the local thermal non-equilibrium (LTNE) model more suitable to describe the heat transfer behavior than the local thermal equilibrium (LTE) model [5,6]. For instance, experimental observations by Zheng et al. [7] and Song et al. [8] confirmed the LTNE mechanism for porous methane combustion and biomass combustion due to the significant differences between gas temperatures (T_g) and solid temperatures (T_s). Zannoni et al. [9] demonstrated that the LTNE model was essential for the accurate prediction of low- Re smoldering

combustion inside porous media. In the LTNE model, the gas-solid interfacial heat transfer coefficient (h_{sg}) is a critical parameter coupling the gas-and-solid energy conservation equations, as shown in Eqs. (1) and (2).

$$\varepsilon(\rho_g C_{pg}) \frac{\partial T_g}{\partial t} + \rho_g C_{pg} u_g \frac{\partial T_g}{\partial x} = \varepsilon \frac{\partial}{\partial x} \left(k_g \frac{\partial T_g}{\partial x} \right) + h_{sg} \left(\frac{A_{s,sp}}{V_{sp}} \right) (T_s - T_g) \quad (1)$$

$$(1 - \varepsilon)(\rho_s C_{ps}) \frac{\partial T_s}{\partial t} = (1 - \varepsilon) \frac{\partial}{\partial x} \left((k_s + k_{rad}) \frac{\partial T_s}{\partial x} \right) - U \left(\frac{A_{s,c}}{V_c} \right) (T_s - T_\infty) + h_{sg} \left(\frac{A_{s,sp}}{V_{sp}} \right) (T_g - T_s) \quad (2)$$

An empirical correlation proposed by Wakao et al. [10] ($Nu = 2 + 1.1(Re^{0.6}Pr^{1/3})$, $15 \leq Re \leq 8500$) has been often employed to estimate h_{sg} in porous media. However, it has been revealed that this correlation would tend to overestimate h_{sg} under low- Re condition ($Re \leq O(10^2)$) [11–14]. Pioneering work has been undertaken to estimate h_{sg} for porous media [13–21]. Empirical correlations for Nusselt (Nu) number with respect to Re and Pr numbers are summarized in Table 1.

Since experimental temperature data inside porous media was not integrity and only T_s was available, there were two unknown variables, i.e. h_{sg} and T_g and then two energy conservation equations i.e. the LTNE

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Nomenclatures		Greek Symbols	
<i>Latin Letters</i>		ε	porosity
C_p	specific heat capacity, J kg ⁻¹ K ⁻¹	ρ	density, kg m ⁻³
d	particle size, mm	Δ	increment
h_{sg}	gas-solid interfacial heat transfer coefficient, W m ⁻² K ⁻¹	<i>Subscripts</i>	
k	thermal conductivity, W m ⁻¹ K ⁻¹	a	ambient
k_{rad}	radiative thermal conductivity, W m ⁻¹ K ⁻¹	c	composite
NRMSD	Normalized Root Mean Square Deviation, -	g	gas phase
t	time, h	i	time
T	temperature, K	j	space
u	Darcy flux, m s ⁻¹	s	solid phase
U	global energy loss coefficient, W m ⁻² K ⁻¹	p	particle
V	volume, m ³	∞	environment
x	axial height, m		

model shown in Eqs. (1) and (2) had to be solved. It is an ill-posed problem for solving h_{sg} from the LTNE model and no analytical solution exists. The sole solution in the literature as shown in Table 1, was to develop the inverse computation algorithms for optimizing h_{sg} while the differences between experimental and numerical temperatures were minimized [7,13–16,18–21], in which the most typical approach was called a ‘transient single-blow technique’ [13,18,19,22,23]. This technique measured T_s inside porous media and inlet and outlet ambient temperature (T_a) of porous media, and then h_{sg} was optimized by minimizing the experimental and numerical outlet T_a . However, the radial heat loss for the transient single-blow technique was ignored. It was a critical issue that could lead to a considerable error to the estimate of h_{sg} because the radial heat loss was remarkable as elucidated in recent work [24–26]. This critical issue has been recently addressed by Zanoni et al. [15], in which the radial heat loss was incorporated in the solid energy conservation equation (i.e. U in Eq. (2)) and estimated by the global energy balance.

There are two critical issues faced by the inverse computation for h_{sg} : (i) high computation cost and (ii) numerical uncertainty resulting from both numerical computation of the governing equations and inverse optimization. These issues are mainly caused by the inaccessible integrity data of both T_s and T_g inside porous media. Supposing data is integrity and both T_s and T_g can be accessible from experiments, then h_{sg} becomes the sole unknown parameter and it could be analytically solved from only one energy conservation equation. Thus, the data integrity could be a novel solution to determine h_{sg} and very promising to address these issues by reducing the computation cost and numerical uncertainty.

This work aims to resolve h_{sg} under low- Re conditions ($13 \leq Re \leq 133$) using the data integrity of both T_s and T_g measured by our newly developed experimental measuring technique [8]. The experimental setup for simultaneous measurement of T_s and T_g is firstly introduced. Four variables including Darcy flux (volume per unit cross sectional area per unit time [15], also named the filtration velocity), particle size, porous medium and heating temperature are considered in experiments.

With the data integrity of both T_s and T_g available from experiments, h_{sg} is directly solved from the gas energy conservation equation and approximated with the central difference scheme. An empirical correlation of Nu with respect to Re and Pr is established based on 54 studied cases. Sensitivity analysis for the analytical solution of h_{sg} with the data integrity is conducted. Comparisons to other empirical correlations in the literature and limitations of this study are discussed. This work develops a promising approach from a new perspective of data integrity to resolve the challenges faced by the inverse computation for h_{sg} in porous media.

2. Methodology

2.1. Experiment setup

The schematic diagram of the experimental setup was depicted in Fig. 1. The geometric parameter of the reactor was with the radius of 12 cm and the height of 60 cm. The air supply system delivered constant-pressure gas via an air compressor, with flow rates controlled through the mass flow controller (MFC-700, DESIGN TECH, China). A 3 cm layer of coarse rock particles was placed at the bottom to ensure uniform gas flow. A heating coil was placed at the bottom of reactor to heat porous media to a programmed temperature. Two sets of K-type TCs were inserted with 2.6 cm intervals into the central axis of the reactor. TC data was recorded by a data acquisition instrument (U2331, KEYSIGHT TECHNOLOGIES, USA) with a 4 second time resolution. As can be seen in Fig. 1, the coated TCs at the left-hand side of the reactor were used to measure T_s at the central axis of the reactor, whereas the coated TCs embedded inside porous tubes at the opposite side were employed to measure T_g at the central axis of the reactor. A hollowed space inside a porous tube was created for measuring T_g . To alleviate the perturbation induced by invasive measurement, the hollowed porous tube mimicked the porous property of porous media in the reactor through peripherally opening several holes in the tube (see Fig. 1), which allowed gas flow through the tube and meanwhile controlled the porosity of the tube

Table 1
Summary of Nu empirical correlations for porous media under $Re \leq O(10^2)$ conditions.

Nu correlations	Re	Porous medium	Porosity	Data	Method	Ref.
$Nu = 0.001Re^{1.97}Pr^{1/3}$	0.5-31	Quartz sand	0.36-0.39	T_s	Inverse computation	[15]
$Nu = 0.004Re^{1.35}Pr^{1/3}$	0.5-10	Powder metal	0.054	T_s, T_a	Inverse computation	[27]
$Nu = 0.025Re^{0.97}Pr^{1/3}$	20-200	Powder metal	0.26-0.653	T_s, T_a	Inverse computation	[27]
$Nu = 0.048Re^{0.7}Pr^{1/3}$	20-100	Granular potash	-	T_s, T_a	Inverse computation	[28]
$Nu = 0.057Re^{0.96}Pr^{1/3}$	20-500	Low carbon steel	0.23-0.39	T_a	Inverse computation	[29]
$Nu = 2.74 + 3.08Re^{0.2}Pr^{1/3}$	17-57	Metal	0.32	T_s, T_a	Approximate solution	[30]
$Nu = 0.049Re^{0.857}Pr^{1/3}$	5-280	Spherical glass	0.58-0.67	T_s, T_a	Approximate solution	[17]
$Nu = 0.001Re^{1.66}Pr^{1/3}$	13-133	Stainless steel and Quartz sand	0.41-0.49	T_s, T_g	Approximate solution	This work

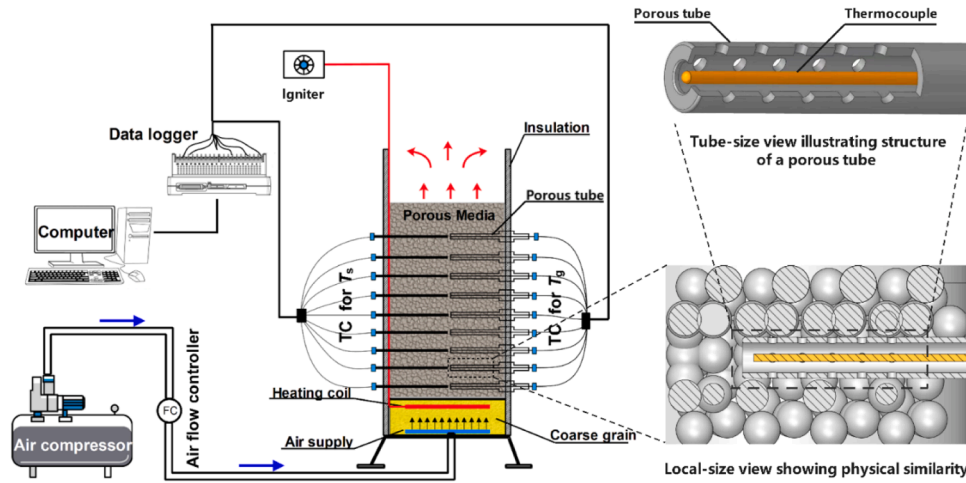


Fig. 1. Schematic diagram of experimental setup.

approximate to that of porous media in the reactor. Details in terms of simultaneous measurement of its uncertainty can be found in [8].

Four variables, including the porous media (quartz sand and stainless steel), particle size (5, 10 and 15 mm), Darcy flux (1.47, 2.95 and 5.89 $\text{cm}\cdot\text{s}^{-1}$) and heating temperature (300, 400 and 500°C), were considered in the experiments. Therefore, 54 tests in total were conducted in this study, as listed in Table 2. Each experiment was conducted at least twice to verify experimental repeatability.

2.2. Approximate solution

Since both T_s and T_g were obtained, the h_{sg} became the only unknown parameter in the energy conservation equations. Consequently, h_{sg} can be directly solved using just one energy conservation equation. In this study, the gas phase energy conservation equation shown in Eq. (2)

was considered, as it included u_g , enabling comprehensive evaluation of the h_{sg} , and excluded the extra parameter U in Eq. (1) reducing uncertainty.

The gas energy conservation equation was discretized using the central difference scheme. It was assumed that solid particles were spherical, and their surface area per unit volume was $6(1-\epsilon)/d_p$ [31]. Then, h_{sg} was approximately solved as follows.

$$h_{sg} = \left(\epsilon \rho_g C_{pg} \frac{T_g(x, t + \Delta t) - T_g(x, t)}{\Delta t} + \rho_g C_{pg} u_g \frac{T_g(x + \Delta x, t) - T_g(x, t)}{\Delta x} - \epsilon k_g \frac{T_g(x + \Delta x, t) - 2T_g(x, t) + T_g(x - \Delta x, t)}{\Delta x^2} \right) \frac{d_p}{6(1 - \epsilon)(T_s(x, t) - T_g(x, t))} \quad (3)$$

where $\Delta x = 2.6$ cm denoting the vertical interval between adjacent thermocouple pairs as shown in Fig. 1, and $\Delta t = 4$ s that was the data acquisition resolution of the thermocouples.

2.3. Mean value calculation

The gas-solid interfacial heat transfer coefficient, h_{sg} , was a parameter varying with heat transfer process. As can be seen from Eq. (3), it varied with time and in space. Harmonic mean was a type of numerical average that was usually used in situations when the average rate or rate of change needed to be calculated. It was usually used to find the average of rates as it assigned equal weight to all data points in a sample. h_{sg} can be considered as a rate-type parameter since $[W] = [J\cdot s^{-1}]$. Thus, the harmonic mean as shown in Eq. (4) was calculated to represent the characteristic value for establishing the empirical correlation of Nu .

$$\bar{h}_{sg} = \frac{N}{\sum_{i=1}^N \frac{1}{h_{sg,i}}} \quad (4)$$

2.4. Local sensitivity analysis

It was reported in our previous work that the measurement of T_g was the key uncertainty for the determination of h_{sg} [8]. The sensitivity of h_{sg} with regarding to T_g measurement uncertainty was calculated by the following logarithmic sensitivity coefficient.

$$S = \left| \frac{\sum_{j=1}^{M-2} \sum_{i=1}^N (\log h'_{sg,ij} - \log h_{sg,ij}^{\exp})}{\sum_{j=1}^M \sum_{i=1}^N (\log T_{g,ij} - \log T_{g,ij}^{\exp})} \right| \quad (5)$$

Table 2

Summaries of studied cases and approximate solution of h_{sg} .

d_p/mm	$u_g/\text{cm}\cdot\text{s}^{-1}$	Heating temperature/°C	Quartz sand $\bar{h}_{sg}/\text{W}\cdot\text{m}^{-2}\cdot\text{K}^{-1}$	304 stainless steel $\bar{h}_{sg}/\text{W}\cdot\text{m}^{-2}\cdot\text{K}^{-1}$
5	1.47	300	0.030	0.029
		400	0.032	0.110
		500	0.019	0.089
	2.95	300	0.527	0.232
		400	0.265	0.130
		500	0.274	0.213
	5.89	300	2.013	2.794
		400	1.968	1.943
		500	2.524	1.481
10	1.47	300	0.240	0.278
		400	0.015	0.266
		500	0.054	0.128
	2.95	300	0.830	0.632
		400	0.437	1.068
		500	0.301	0.717
	5.89	300	6.304	3.493
		400	6.177	5.734
		500	4.169	4.823
15	1.47	300	0.127	0.848
		400	0.397	0.793
		500	0.378	0.703
	2.95	300	1.452	1.122
		400	1.797	0.373
		500	1.574	0.653
	5.89	300	5.683	6.058
		400	6.480	5.689
		500	5.266	5.714

where the superscript ' stood for the terms with uncertainty, and the subscripts i and j denoted time and space, respectively. The uncertainty of T_g measurement was quantified with *NRMSD* as illustrated in our previous work [8]. Both quartz sand and stainless steel studied porous media were considered and their *NRMSD* varied 1.56% - 1.69% and 1.28% - 2.32%, respectively. The plots of T_g history data with measurement uncertainty comparing to experimental data were presented in Fig. 2.

3. Results and discussion

The harmonic mean $\bar{h}_{sg}$ under varied experimental conditions are obtained as shown in Table 2. The *Nu* number of the convective heat transfer inside the porous media is obtained by Eq. (6) in the following with $\bar{h}_{sg}$ listed in Table 2.

$$Nu = \frac{\bar{h}_{sg} d_p}{k_g} \quad (6)$$

Reynolds number ($Re = \rho_g u_g d_p / \mu_g$) and Prandtl number ($Pr = \mu_g C_{pg} / k_g$) are calculated for all studied cases with $k_g = 0.028 \text{ W m}^{-1} \text{ K}^{-1}$, $\rho_g = 1.29 \text{ kg m}^{-3}$, $\mu_g = 1.75 \times 10^{-5} \text{ kg m}^{-1} \text{ s}^{-1}$, $C_{pg} = 1050 \text{ J kg}^{-1} \text{ K}^{-1}$, and $\varepsilon = 0.41 - 0.49$ depending on particle size [8]. Then, the Levenberg-Marquardt (LM) algorithm is used to establish an empirical correlation of *Nu* with respect to *Re* and *Pr*, as shown in Eq. (7).

$$Nu = 0.001 Re^{1.66} Pr^{1/3} \quad (R^2 = 0.93) \quad (7)$$

Figure 3 compares the correlation in this work to three *Nu* empirical correlations by Zani et al. [15], Nie et al. [16], and Wakao et al. [10]. Other *Nu* empirical correlations under low-*Re* condition ($Re \leq 10^2$) are summarized in Table 1. The magnitude of *Nu* resolved by data integrity of T_s and T_g in this work is close to Nie et al. [17] and Zani et al. [11], which is much lower than that extrapolated by Wakao et al. [8]. It confirms the research findings in [7,11] that the gas-solid interfacial heat transfer mechanism changes significantly and local thermal nonequilibrium would emerge while *Re* decreases to below 100. Comparisons of *Nu* empirical correlations shown in Fig. 3 indicates that the correlation established in this work is in line with previous work such as Nie et al. [17] and Zani et al. [11]. It demonstrates the proposed approach with the data integrity of both T_s and T_g owns a good capability for the determination of h_{sg} . Furthermore, taking advantages of the analytical solution of h_{sg} from the gas energy conservation equation, this approach is more efficient by saving enormous numerical computations and inverse optimizations. Besides, the proposed approach avoids the inverse computation errors because it does not involve any numerical computations and inverse optimizations.

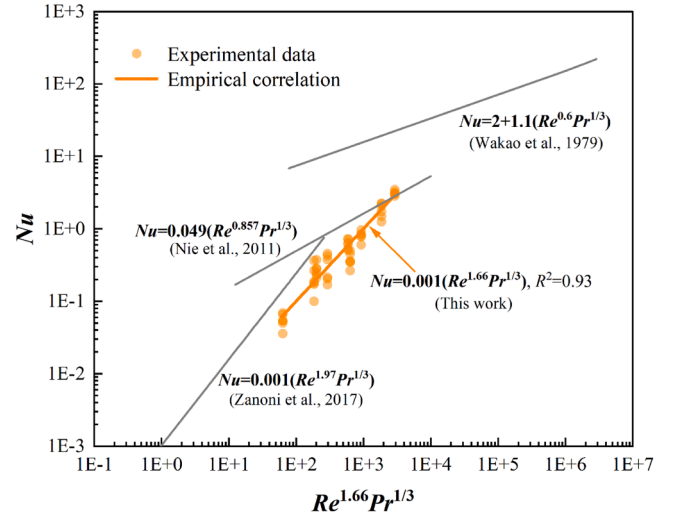


Fig. 3. Comparisons of *Nu* empirical correlation in this work with Zani et al. [15], Nie et al. [17], and Wakao et al. [10].

The approximate solution of h_{sg} in this work is slightly smaller ($O(10^{-1})$) than the approximate solution of h_{sg} in [17] and inverse computation for h_{sg} with the radial heat loss in [15]. Such slight diverges could be attributed to measurement noises of T_g in this work, especially for quartz sand [8], and also associated with errors of approximate solution in [17] and inverse computation in [15]. For instance, in [17], only one solid energy conservation equation was solved for h_{sg} , in which the other unknown variable T_g inside porous media was replaced by the ambient temperature (T_a). Nevertheless, the *Nu* empirical correlation in this work extrapolated towards larger *Re* number and lower *Re* number could merge with the correlations by Nie et al. [17] and Zani et al. [15]. Note that the Darcy flux in practice such as the filtration combustion could be high to $\sim 0.24 \text{ m s}^{-1}$ [15,16]. The main reason for the small velocities was attributed to the comparisons with the previous work [15,17] validating the proposed approach in this work. The applicability of the established correlation in this work extending to larger Darcy flux will be examined in the future.

The proposed approach is mainly limited by two uncertainties: experimental uncertainty and numerical uncertainty. As for the experimental uncertainty, Fig. 4 shows the sensitivity of h_{sg} to the T_g measurement uncertainty. The T_g measurement error resulting from the porous tubes could lead to significant bias of the estimate of h_{sg} . It indicates that attention must be paid to reduce the T_g measurement error, for instance, optimizing experimental operations and improving

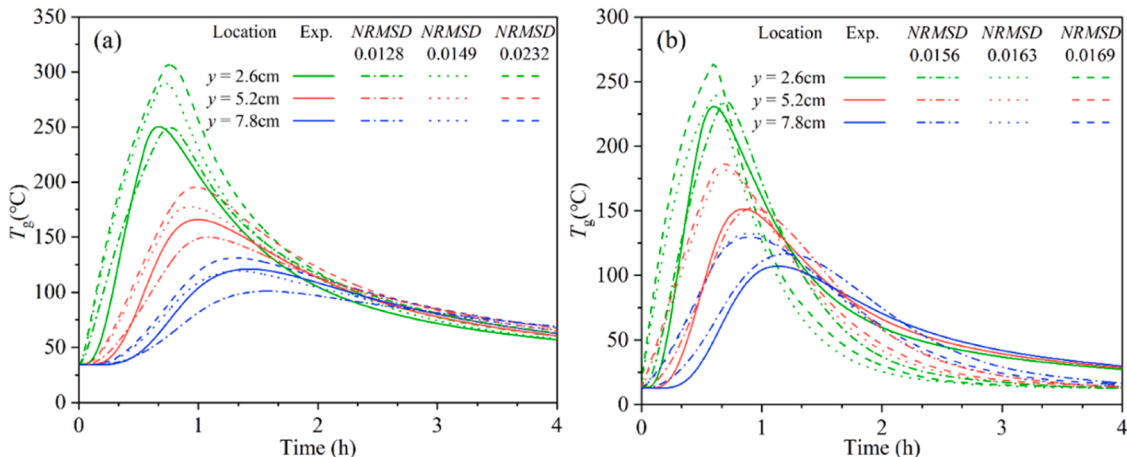


Fig. 2. Plots of T_g history data with measurement uncertainty comparing to experimental data (a) stainless steel, and (b) quartz sand.

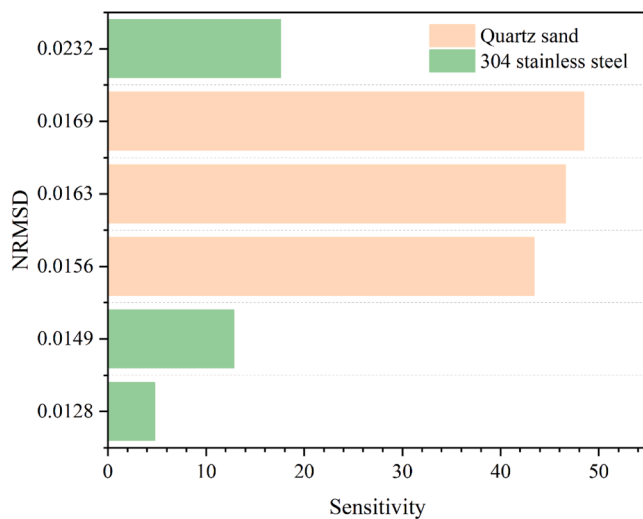


Fig. 4. Sensitivity analysis for T_g measurement uncertainty.

measuring techniques for T_g , h_{sg} is more sensitive to quartz sand than stainless steel. This could be mainly attributed to the thermal heterogeneity among the porous stainless-steel tubes and quartz sand medium [8]. It suggests that the confidence of h_{sg} solved by Eq. (3) for quartz sand could be lower than stainless steel. h_{sg} for quartz sand could bring more noise to the established Nu empirical correlation.

The numerical uncertainty is associated with the discretization errors of the central difference scheme, which depends on the measuring intervals at the central axis of reactor. It is expected that a finer interval in experiments could decrease the discretization errors. Besides, the discretization issue could be solved by mesh-free numerical algorithms, such as physics-informed neural networks (PINNs) [32] that incorporate fundamental physical principles as soft constraints in model training, thereby ensuring that all solutions rigorously adhere to the underlying governing equations. The advantage of PINNs over the finite difference methods lies in the mesh free, which does not require a predefined grid, potentially providing a continuous solution field even with sparse or irregularly spaced data points. Then, the numerical uncertainty associated with the discretization errors of the central difference scheme could be avoided. This hybrid data-physics framework demonstrates superior capability in characterizing complex heat transfer phenomena [33], which might be helpful to eliminate the numerical uncertainty of the proposed method using data integrity to resolve h_{sg} in the future.

4. Conclusions

Data integrity achieved by simultaneous measurements of T_s and T_g in porous media is implemented in a discretized gas energy conservation equation to approximately solve h_{sg} in porous media for $5 \text{ mm} \leq d_p \leq 15 \text{ mm}$ and $13 \leq Re \leq 133$. This solution takes advantages in simplifying computation with only one equation instead of two energy conservation equations, and reducing uncertainty with direct heat-transfer information from local T_s and T_g data rather than deduction of h_{sg} only from T_s data. An empirical correlation of $Nu = 0.001(Re^{1.66}Pr^{1/3})$ is derived, which is in accordance with other correlations developed in the literature. It demonstrates that the data integrity provides a new perspective to resolve challenges in the inverse models of heat transfer in porous media.

CRediT authorship contribution statement

Zeyang Song: Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Jian Wang:** Writing – original

draft, Investigation, Formal analysis. **Shaotang Hui:** Methodology, Investigation, Formal analysis, Data curation. **Renkun Dai:** Writing – review & editing, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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